



## **Develop your approach to cleaning data**

As you continue on your data journey, you're likely discovering that data is often messy—and you can expect raw, primary data to be imperfect. In this reading, you'll consider how to develop your personal approach to cleaning data. You will explore the idea of a cleaning checklist, which you can use to guide your cleaning process. Then, you'll define your preferred methods for cleaning data. By the time you complete this reading, you'll have a better understanding of how to methodically approach the data cleaning process. This will save you time when cleaning data and help you ensure that your data is clean and usable.

### **Consider your approach to cleaning data**

Data cleaning usually requires a lot of time, energy, and attention. But there are two steps you can take before you begin to help streamline your process: creating a cleaning checklist and deciding on your preferred methods. This will help ensure that you know exactly how you want to approach data cleaning and what you need to do to be confident in the integrity of your data.

#### **Your cleaning checklist**

Start developing your personal approach to cleaning data by creating a checklist to help you identify problems in your data efficiently and identify the scale and scope of your dataset. Think of this checklist as your default “what to search for” list.

Here are some examples of common data cleaning tasks you could include in your checklist:

- Determine the size of the dataset: Large datasets may have more data quality issues and take longer to process. This may impact your choice of data cleaning techniques and how much time to allocate to the project.
- Determine the number of categories or labels: By understanding the number and nature of categories and labels in a dataset, you can better understand the diversity of the dataset. This understanding also helps inform data merging and migration strategies.
- Identify missing data: Recognizing missing data helps you understand data quality so you can take appropriate steps to remediate the problem. Data integrity is important for accurate and unbiased analysis.
- Identify unformatted data: Identifying improperly or inconsistently formatted data helps analysts ensure data uniformity. This is essential for accurate analysis and visualization.
- Explore the different data types: Understanding the types of data in your dataset (for instance, numerical, categorical, text) helps you select appropriate cleaning methods and apply relevant data analysis techniques.

There might be other data cleaning tasks you've been learning about that you also want to prioritize in your checklist. Your checklist is an opportunity for you to define exactly what you want to remember about cleaning your data; feel free to make it your own.

### **Your preferred cleaning methods**

In addition to creating a checklist, identify which actions or tools you prefer using when cleaning data. You'll use these tools and techniques with each new dataset—or whenever you encounter issues in a dataset—so this list should be compatible with your checklist.

For example, suppose you have a large dataset with missing data. You'll want to know how to check for missing data in larger datasets, and how you plan to handle any

missing data, before you start cleaning. Outlining your preferred methods can save you lots of time and energy.

## **Key takeaways**

The data you encounter as an analyst won't always conform to your checklist or your preferred actions and tools. But having these things can make common data cleaning tasks much easier to complete. As is so often the case, thoughtful planning sets up any project for success!

### **Key Concepts:**

- \* Data Quality Issues: Problems in the dataset that can affect the integrity and usability of the data, such as missing or unformatted data.
- \* Dataset Size and Diversity: Understanding the size and nature of a dataset, which informs the choice of cleaning techniques and strategies.
- \* Preferred Cleaning Methods: The specific actions or tools an analyst prefers to use when cleaning data, which can streamline the cleaning process.
- \* Data Cleaning Checklist: A personalized list of common data cleaning tasks that helps identify problems in a dataset efficiently.